

# SemanticCueSync at AraIEval Shared Task: Detecting Propagandistic Spans with Persuasion Techniques Identification using Pre-trained Transformers

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## Abstract

Detecting propagandistic spans and identifying persuasion techniques are crucial for promoting informed decision-making, safeguarding democratic processes, and fostering a media environment characterized by integrity and transparency. Various machine learning (Logistic Regression, Random Forest, and Multinomial Naive Bayes), deep learning (CNN, CNN+LSTM, CNN+BiLSTM), and transformer-based (AraBERTv2, AraBERT-NER, CamelBERT, BERT-Base-Arab) models were exploited to perform the task. The evaluation results indicate that CamelBERT achieved the highest micro-F1 score (24.09%), outperforming CNN+LSTM and AraBERTv2. The study found that most models struggle to detect propagandistic spans when multiple spans are present within the same article. Overall, the model's performance secured a 6<sup>th</sup> place ranking in the AraIEval Shared Task-1.

## 1 Introduction

Detecting propagandistic spans with persuasion techniques identification involves pinpointing particular text sections characterized by the deliberate use of persuasive language and tactics to shape opinions, attitudes, or actions. This involves analyzing linguistic cues, rhetorical strategies, and contextual elements to identify instances of propaganda and the specific persuasion techniques employed. It is difficult for a naive user to determine if the information is factually correct or if it is just another propaganda technique. There have been many studies done on detecting fake news (Nguyen et al., 2020), biased (Baly et al., 2020), hyperpartisan (Potthast et al., 2017), and propagandistic news and articles (Da San Martino et al., 2019). The influence of fake news on Twitter (Bovet and Makse, 2019) and other online media (Vosoughi et al., 2018) can be dangerous. Deliberately spreading fake news about a particular topic (Nakov et al.,

2021a,b) is also a propaganda technique. Social media users tend to believe anything with more views without any factual basis (Guo et al., 2020).

Propaganda detection (Pastor et al., 2024) is a multidisciplinary field that draws upon techniques from NLP and identifies elements indicative of propaganda. There are diverse propaganda techniques (Jones, 2024), and classifying them is a challenge. Span detection (Li et al., 2022; Martino et al., 2020) in propaganda techniques is another big challenge. Propaganda aims to influence the audience to advance a specific agenda. Detecting the exact span where it occurs is tricky and arguably more difficult than finding false information in a news article. Few studies have focused on propaganda technique detection (Othman, 2023; Refaee et al., 2022) and span detection (Attieh and Hassan, 2022) in Arabic.

By applying propagandistic textual spans detection with persuasion techniques identification across various domains, individuals and organizations can better understand and counteract the influence of propaganda, ultimately fostering more informed, ethical, and resilient societies. The contributions of this work are illustrated as follows:

- Proposed a fine-tuned transformer model to classify various propaganda techniques and detect the span of propaganda techniques.
- Investigated several ML, DL, and transformer-based models for detecting propagandistic techniques and performed in-depth error analysis, offering valuable insights into detecting propagandistic span and persuasion techniques.

## 2 Related Work

Propaganda detection has been the subject of numerous research in recent years. However, the majority of them were primarily concerned with English. Li et al. (2019) proposed a model to detect propaganda at the sentence level using logistic

regression, which achieved an  $f_1$ -score of 0.66. Several DL, ML, and transformer-based techniques were proposed by Gupta et al. (2019) to detect propaganda. The ensemble technique yielded the highest  $f_1$ -score of 0.669 out of all of them. Both of these studies approached propaganda detection as a binary classification task.

Propaganda detection in Arabic has seen substantial advancements in recent years. Hasanain et al. (2023b) provided an overview of the ArAIEval shared task, presented at the first ArabicNLP 2023 conference, focusing on persuasion technique identification and disinformation detection in both binary and multitask settings. Alam et al. (2022) organized a shared task specifically on propaganda detection in Arabic Tweets, addressing a wide range of topics. This shared task included two objectives: identifying the type of propaganda technique employed and determining the specific text spans where these techniques were utilized. Refaee et al. (2022) used pre-trained BERT model to detect propaganda in Arabic tweets. Their model achieved an  $f_1$ -score of 0.602. In contrast to the earlier study, Samir et al. (2022) details how to detect propaganda and its validity span. GPT-4 was used in experiments by Hasanain et al. (2024a) to identify and locate the propaganda span on multiple languages’ news datasets. They found that GPT-4’s performance is low for this sequence tagging and multilabel task, especially in low-resourced languages. Hasanain et al. (2023a) proposed using LLM for propaganda span annotation. This work uses a transformer-based model to address the downstream task of detecting propagandistic techniques from Arabic tweets and paragraphs extracted from news articles.

### 3 Task and Dataset Description

To address the phenomena of detecting propagandistic techniques, shared task organizers<sup>1</sup> provided a unimodal (text) dataset. The dataset (Hasanain et al., 2024b) contains Arabic tweets and paragraphs for propagandistic technique detection. The dataset covers two genres: tweets and paragraphs extracted from news articles. The dataset covers a total of 23 different classes representing various propaganda techniques. Table 1 illustrates the distribution of train, dev, and test sets for tweets and paragraphs, where  $T_W$  denotes total words.

The train, dev, and test sets consisted of 6997,

| Type      | Train | Dev | Test | $T_W$ |
|-----------|-------|-----|------|-------|
| Tweet     | 995   | 249 | 260  | 1504  |
| Paragraph | 6002  | 672 | 786  | 7460  |
| Total     | 6997  | 921 | 1046 | 8964  |

Table 1: Dataset statistics for Task-1.

921, and 1046 Arabic tweets and paragraphs extracted from news articles. Task-1 (Hasanain et al., 2024b) focused on detecting and classifying the propaganda techniques used in the text and the exact span(s) where each technique appears. Table 2 illustrates a few examples of text, technique, and the start and end of the span in the dataset. The "start" and "end" columns indicate which part of the text contains propaganda.

| Text                               | Technique             | Start | End |
|------------------------------------|-----------------------|-------|-----|
| حرب (War)                          | Loaded-Language       | 11    | 14  |
| عبيد الطغاة<br>(Slaves of tyrants) | Name-Calling-Labeling | 78    | 89  |
| شبهة (suspicion)                   | Doubt                 | 28    | 32  |
| الإصرار (Determination)            | Loaded-Language       | 52    | 59  |

Table 2: Task-1 sample with text, technique, and start and end of span.

## 4 System Development

Figure 1 illustrates the schematic process of classifying propaganda techniques and detecting textual spans. We used a tokenizer to tokenize each text. We added unique tokens  $[CLS]$  at the beginning and  $[SEP]$  at the end and assigned a  $-100$  label to both so that the loss function ignores them. Then, we used the BIO tagging scheme (Ramshaw and Marcus, 1999; Dai et al., 2019), where the B-tag indicates the beginning of an entity, the I-tag indicates a token contained inside the same entity, and the O-tag denotes that no entity is contained within the token.

### 4.1 Classifiers

We explored several ML, DL, and transformer-based models for this task.

- **ML-based Models:** Logistic Regression (LR), Random Forest (RF), and Multinomial Naive Bayes (MNB) were used for this task.

<sup>1</sup><https://araieval.gitlab.io/task1/>

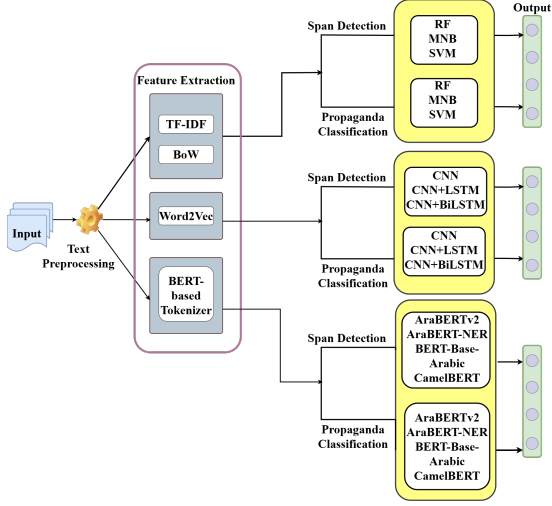


Figure 1: Schematic process for Propagandistic Technique Detection.

| Classifier | Parameters               | Value          |
|------------|--------------------------|----------------|
| RF         | max-depth<br>n-estimator | 20<br>500      |
| MNB        | alpha<br>fit-prior       | 1.0<br>False   |
| LR         | solver<br>max_iter       | lbfgs<br>20000 |

Table 3: Parameters used for ML models

We employed Bag-of-Words (BoW) method for feature extraction using ‘CountVectorizer’ from Scikit-learn<sup>2</sup>. Besides we also used TF-IDF from Scikit-learn for extracting textual features. A maximum depth of 20 was employed with 500 estimators and the ‘*max\_features*’ was set to ‘sqrt’ in the RF classifier. For MNB, we set the alpha value at 0.1 and fit before False. Lastly, the LR is used with the ‘lbfgs’ solver and max iter set to 20000. Table 3 shows the several ML parameters used for the models.

- **DL-based Models:** The CNN model uses an embedding layer with an output dimension of 256. It comprises one Conv1D layer with 512 filters and a GlobalMaxPooling layer for downsampling. To prevent overfitting, non-linearity is introduced by the dense layer with L2 regularisation. The output layer allows for multi-class classification with its 23 units and softmax activation. We also set loss to ‘SparseCategoricalCrossentropy’, optimizer to

‘adam’ and metrics to ‘accuracy’.

**CNN+LSTM:** This method utilized one embedding layer with output-dim=256, one Conv1D layer, one MaxPooling1D layer, three LSTM layers, and three dense layers with a dropout layer. The number of units for the Conv1D layer was 512, whereas the LSTM layers had 512, 256, and 128. The unit size for the first dense layer was 256 with ‘tanh’ activation; the second dense layer had 128 units with ‘relu’ activation; and the third dense layer had 23 units for classification with ‘softmax’ activation. The loss function utilized in this case was ‘SparseCategoricalCrossentropy’ with the ‘Adam’ optimizer.

**CNN+BiLSTM:** This method utilized one Conv1D layer, one MaxPooling1D layer, three Bidirectional LSTM layers, and three dense layers with a dropout layer. The number of units for the Conv1D layer was 512, whereas the LSTM layers had 512, 256, and 128. The unit size for the first dense layer was 256 with ‘relu’ activation; the second dense layer had 128 units with ‘relu’ activation; and the third dense layer had 23 units for classification with ‘softmax’ activation. The loss function utilized in this case was ‘SparseCategoricalCrossentropy’ with the ‘Adam’ optimizer.

| Models         | LR   | WD    | BS | EP |
|----------------|------|-------|----|----|
| AraBERTv2      | 1e-5 | 0.001 | 8  | 5  |
| AraBERT-NER    | 1e-5 | 0     | 32 | 3  |
| BERT-Base-Arab | 5e-5 | 0.001 | 8  | 5  |
| CamelBERT      | 5e-5 | 0.001 | 8  | 6  |

Table 4: Hyperparameters for transformer-based models, where LR, WD, BS, and EP denote learning rate, weight decay, batch size, and epochs, respectively.

- **Transformer-based Models:** BERT (Bidirectional Encoder Representations from Transformers), are developed to extract contextualized word representations from unlabeled texts (Devlin et al., 2018). It utilizes the transformer architecture’s encoder representation approach, initially introduced by Vaswani et al., 2017. Transformer library of Huggingface<sup>3</sup> includes numerous pre-trained models for text processing. In order to determine the propaganda techniques,

<sup>2</sup><https://scikit-learn.org>

<sup>3</sup><https://huggingface.co/docs/transformers/en/index>

we employed CamelBERT (Inoue et al., 2021), AraBERTv2 (Antoun et al., 2020), BERT-Base-Arabic (Safaya et al., 2020) and AraBERT-NER. Table 4 illustrates the hyper-parameters used in these models. This study presents a framework that leverages CamelBERT for embedding model generation and token classification. Initially, the text was tokenized using CamelBERT’s tokenizer. The tokenization process transforms each token into a dense vector representation through the embedding layer of the model. This approach enables the extraction of intricate semantic and syntactic features. In addition, a specialized classification head was incorporated into CamelBERT for token classification. This process ensures accurate and context-aware token-level predictions.

## 5 Results

Table 5 shows the evaluation results on the test set. Results reveal that LR achieved the most el-

| ML Models         |              |              |              |              |
|-------------------|--------------|--------------|--------------|--------------|
| Classifier        | P            | R            | MF1          | mF1          |
| <b>LR</b>         | <b>6.51</b>  | <b>60.37</b> | <b>0.47</b>  | <b>11.76</b> |
| RF                | 6.36         | 61.76        | 0.23         | 11.54        |
| MNB               | 6.27         | 57.25        | 0.57         | 11.31        |
| DL Models         |              |              |              |              |
| Classifier        | P            | R            | MF1          | mF1          |
| CNN               | 6.21         | 60.33        | 0.32         | 11.26        |
| CNN+LSTM          | 6.36         | 61.76        | 0.23         | 11.54        |
| <b>CNN+BiLSTM</b> | <b>6.36</b>  | <b>61.76</b> | <b>0.23</b>  | <b>11.54</b> |
| Transformers      |              |              |              |              |
| Classifier        | P            | R            | MF1          | mF1          |
| AraBERTv2         | 34.81        | 18.12        | 14.43        | 23.84        |
| <b>CamelBERT</b>  | <b>20.54</b> | <b>29.11</b> | <b>12.93</b> | <b>24.09</b> |
| AraBERT-NER       | 5.42         | 13.68        | 0.23         | 7.76         |
| BERT-Base-Arabic  | 23.20        | 18.06        | 10.28        | 20.31        |

Table 5: Performance of the employed models on the test set. This table displays the Precision (P), Recall (R), macro F1 (MF1), and micro F1 (mF1) scores.

evated micro F1-score (11.76%) among the ML approaches, surpassing RF (11.54%) and MNB (11.31%). CNN+LSTM and CNN+BiLSTM with word embeddings have the same micro F1-score of 11.54%, which is approximately 0.22% lower than the best ML approach. On the other hand,

CNN achieved a micro F1 score of 11.26%, slightly slower than the other two approaches. The transformer-based approach performs better on the test dataset. The best-performing transformer model beats the best-performing ML and DL models. CamelBERT, with a micro F1 score of 24.09, is the best performing, while the other three, AraBERTv2, BERT-Base-Arabic, and AraBERT-NER, scored 23.84, 20.31, and 7.7, respectively. The CamelBERT model has produced embeddings that highlight domain-specific features relevant to propaganda. Its tokenizer is superior in handling Arabic morphology and syntax, resulting in more effective tokenization of complex Arabic structures. Consequently, CamelBERT outperforms other models in terms of performance.

### 5.1 Error Analysis

The error analysis with the confusion matrix showed that the employed models can detect Loaded-Language, but the classification of other classes is imperfect. Here, 1680 of the Loaded-Language are correctly classified, and 1067 are incorrectly classified. On the other hand, only 50 of the Name-Calling-Labeling is correctly classified, and the other 254 are incorrectly classified. It is evident that the dataset is significantly unbalanced. 14 out of the 23 classes have no entries in the test set. Furthermore, the Loaded-Language class is overly represented in the test set. The Name-Calling-Labeling class is frequently misclassified as Loaded-Language. Due to these imbalances, the model faces difficulties interpreting the dataset, leading to poor performance on the test set.

Table 6 compares the predicted outcomes and the actual labels. In the first and third samples, the model predicted the propaganda technique accurately but failed to detect the span correctly. In the second example, the model incorrectly predicted the propaganda technique and the span. The test set presented poses significant challenges due to the texts’ extensive length and various propaganda techniques within a single text.

## 6 Discussion

Transformer-based models are capable of identifying various propaganda techniques within the exact text. However, they need to detect the spans consistently and accurately. This inconsistency contributed to the test set’s less-than-optimal perfor-

| Value            | Actual                           | Predicted                  |
|------------------|----------------------------------|----------------------------|
| <b>Technique</b> | Loaded-Language                  | Loaded-Language            |
| <b>Text</b>      | حادث (Accident)                  | اللا (No)                  |
| <b>Start</b>     | 96                               | 0                          |
| <b>End</b>       | 100                              | 4                          |
| <b>Technique</b> | Name-Calling-Labeling            | Questioning-the-Reputation |
| <b>Text</b>      | هجوم تخريبي (Destructive Attack) | أوك (Okay)                 |
| <b>Start</b>     | 192                              | 229                        |
| <b>End</b>       | 203                              | 232                        |
| <b>Technique</b> | Loaded-Language                  | Loaded-Language            |
| <b>Text</b>      | هجوم تخريبي (Destructive Attack) | هجوم (Attack)              |
| <b>Start</b>     | 192                              | 192                        |
| <b>End</b>       | 203                              | 195                        |

Table 6: Few predictions by CamelBERT.

mance. The error analysis findings suggest that most models struggle to detect propaganda span when there are multiple ones in the same article. Most of the errors were introduced in the model span detection phase. Moreover, detecting the span of the propaganda involves the model understanding not only the context but also the tone, intent, and bias. Though the pre-trained BERT-based models can understand contextual elements, they need help in other aspects.

## 7 Conclusion

This study investigates the capabilities of transformer-based models in thoroughly analyzing propaganda span detection and identifying the persuasive techniques in the Arabic dataset. Among all employed ML, DL, and transformer-based models, CamelBERT showed notable performance with the highest micro-F1 score of 24.09. Further advancements could be achieved by expanding the dataset and investigating large language models in the task.

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